

Monitoring and Assessment of Carbon stock through Geospatial Technique in Sathyamangalam Forest, Erode district, Tamilnadu

Giridharan N and Sivakumar R*

Department of Civil Engineering, College of Engineering and Technology,
SRM Institute of Science and Technology, Kattankulathur, Chengalpattu, Tamilnadu, India-603203
Email: gn9372@srmist.edu.in *Corresponding author Email: sivakumr@srmist.edu.in

ABSTRACT

Forest plays a major role in the global carbon balance as both carbon sources and sinks. When considering the greenhouse effect and mitigating climate change, India's top priority for the twenty-first century will likely be managing carbon in the forests. In recent years, Geospatial Technology is employed to monitor and assess the carbon stock in the forest area. It makes very convenient and reduces the cost of estimating carbon storage in forest areas through high-resolution satellite imagery. The present study was analyzed through Non-destructive method i.e. Normalized Difference Vegetation Index (NDVI) is considered a dependent variable and the rest of the indices FPAR, APAR and NPP values are considered independent variables to determine the carbon stock. The results state that the changes in carbon stock with 194.11t/ha (2016) and 144.94 t/ha (2024) are based on climatic actions. The outcome indicates that future development of forest requires continuous monitoring and assessment.

Index Terms— Forest, NDVI, APAR, NPP, carbon stock

1. INTRODUCTION

The forest ecosystem plays an important role in the carbon cycle. A growing number of states are required to create climate action plans, and forest carbon plays a significant role in these plans [1]. Current data on forest carbon stocks and rates of change at pertinent spatial scales is necessary for researchers and other practitioners to create suitable policies and management plans [2]. The plantations that engage in photosynthesis are known to absorb CO₂ from the atmosphere through a process called Carbon sequestration [3]. The Gross primary productivity is determined by the quantity of CO₂ absorbed by the plantations [4]. In addition, the forest serves as a "carbon source" by releasing carbon into the atmosphere as a result of activities like respiration [5]. In respiration, half of the GPP calculates net primary productivity (NPP), or the total amount of biomass matter produced (IPCC 2006). The Forest serves as a source of carbon by raising above-ground biomass respective to increased canopy cover and of soil organic carbon content

[6]. The research on carbon storage contributes to global climate change mitigation, environmental preservation, and the reduction of greenhouse gas emissions in addition to providing strategies for sustainable forest development and management [7].

2. STUDY AREA

The Sathyamangalam Forest (Fig. 1) is located in the Northern part of the Erode district and encloses southwestern part of Tamilnadu. It covers 2438.57 sq. km which includes 199.28 sq. km of non-forest areas such as settlement areas, rocky areas, and waterbodies. Therefore, the forest area is about 2239.29 sq. km. Some of the settlement regions that could be seen between the forest area were described as tribal villages. The forests have a very diverse topography that includes plains, hills, slopes, and rivers. The forest area ranges from 195m and 1739m above mean sea level.

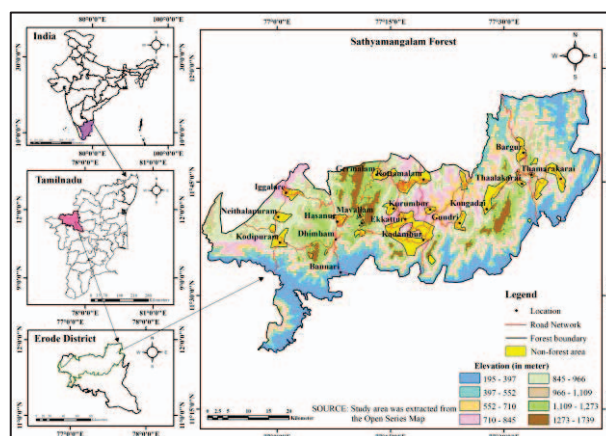


Figure 1. Study area

3. DATA DESCRIPTION

The different indices were derived from SENTINEL 2A for the year 2016 to 2024 (2 years interval) which was obtained from the Bhoonidhi site. It was discovered that two tiles (T43PGP and T43PGN) of SENTINEL 2A data encompassed the study area. In order to carry out the extraction of the study area, the data were mosaiced together.

The Table 1 enumerates the satellite data acquisition for the present study.

Table 1 Data description	
Satellite	Date of Acquisition
SENTINEL 2A	20-03-2016
	22-03-2018
	03-04-2020
	24-03-2022
	36-03-2024

4. METHODOLOGY

The methodology primarily focuses on monitoring and assessing the Sathyamangalam forest's carbon stock. First, the data is collected from the Bhoonidhi website for the various years 2016, 2018, 2020, 2022 and 2024. The Extraction of study area was carried out through QGIS 3.32 software. The various vegetation indices are used to determine the carbon stock estimation such as NDVI, FPAR, APAR and NPP. Values of NDVI and FPAR for various years were estimated from the prepared maps. Similarly, APAR value was calculated using Global solar radiation value of the study area. This APAR value is mandatory to calculate the NPP value.

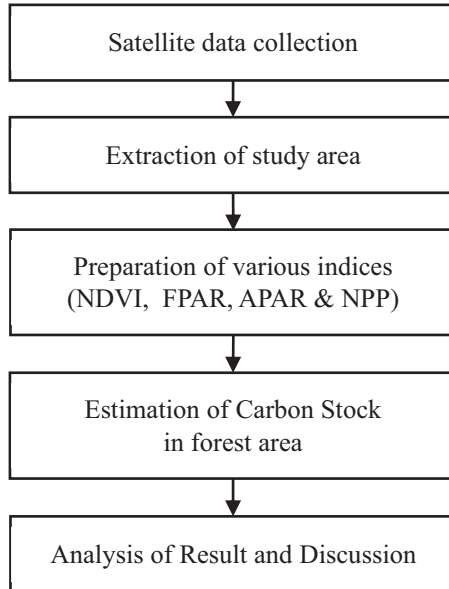


Figure 2. Methodology Flowchart

The carbon stock map was prepared based on the estimated NPP value using QGIS software. The values obtained from the maps are used for quantification purposes. Analysis of the result is carried out and validated with Forest Survey of India carbon stock data. The detailed Methodology flowchart is shown in Figure 2.

4.1 NDVI

The Normalized Difference Vegetation Index (NDVI) accurately indicates the nature of the vegetation in a given region. The NDVI standard ranges between -1 to $+1$ [8]. The positive values signify better vegetation, while the negative values represent less vegetation.

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (1)$$

4.2 FPAR

The amount of solar rays that a plantation absorbs during photosynthesis is known as the Fraction of Absorbed Photosynthetically Active Radiation (FPAR) [9]. It plays a significant role in estimating the carbon stock and serves as an obvious sign of photosynthesis in plantation [10]. The FPAR Index is computed based on its correlation with the NDVI Index as articulated in Equation 2.

$$FPAR = -c + (d * NDVI) \quad (2)$$

The empirical coefficients c and d are -0.08 and 1.075 , in the South Asian region respectively [10].

4.3 APAR

The term Absorbed Photosynthetically Active Radiation (APAR) refers to the total amount of incoming solar radiation (SOL) that the plantation absorbs between 400nm and 700nm [11]. Using the FPAR component and total solar radiation (SOL), the APAR value is estimated (Equation 3).

$$APAR = 0.5 * FPAR * SOL \quad (3)$$

4.4 NPP

The amount of carbon absorbed from the atmosphere during photosynthesis per unit of time is known as net primary productivity [12]. It represents the ecosystem's initial intake of atmospheric carbon and is a crucial indicator for calculating carbon stock [13]. The Equation 4 represents the formula for calculating NPP.

$$NPP = APAR * LUE \quad (4)$$

Light Use Efficiency (LUE) measures the efficiency of plantation which can use incident light for photosynthesis. The LUE for forest is taken as 0.985 [7]

4.5 Carbon stock

The Carbon stock is the quantity of carbon stored in plantations [14]. The estimation of carbon stock was derived with the help of NPP which is given in Equation 5

$$\text{Carbon Stock} = \text{NPP} * 3.67 \quad (5)$$

Consequently, using the features of various indices, it is possible to evaluate the changes in the forest carbon stock values for different years. The present study's methodology is outlined in Figure 2.

5. RESULTS AND DISCUSSION

The Figures 3 to 7 represent the distribution of Carbon stock for 2016, 2018, 2020, 2022 and 2024 respectively. These results confirm changes in carbon stock during the various years based on climatic actions. According to the present study, the carbon stock changed from the minimum to the maximum external region by 194.11 t/ha (2016), 161.64 t/ha (2018), 245.85 t/ha (2020), 183.39 t/ha (2022), and 144.94 t/ha (2024) respectively. The carbon stock increased rapidly in 2020 and was significantly lower in 2018 due to certain environmental actions. The reflectivity of plantations was noticed high in 2020 and 2022 based on high NDVI values due to better climatic conditions as shown in Figure 5 and 6. The Better climatic conditions in 2020 and 2022 led to higher NDVI values, which indicated higher reflectivity of plantations as seen in Figures 5 and 6.

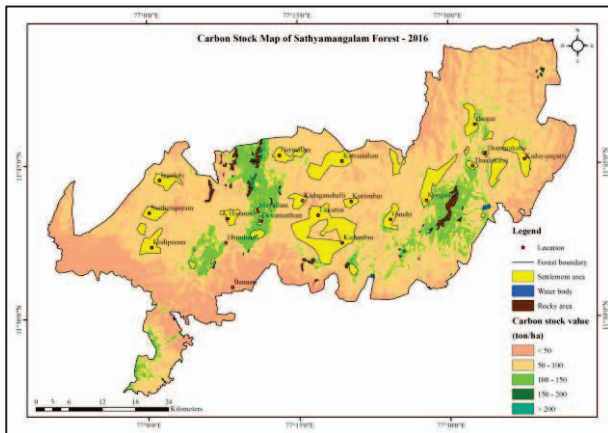


Figure 3. Carbon Stock map – 2016

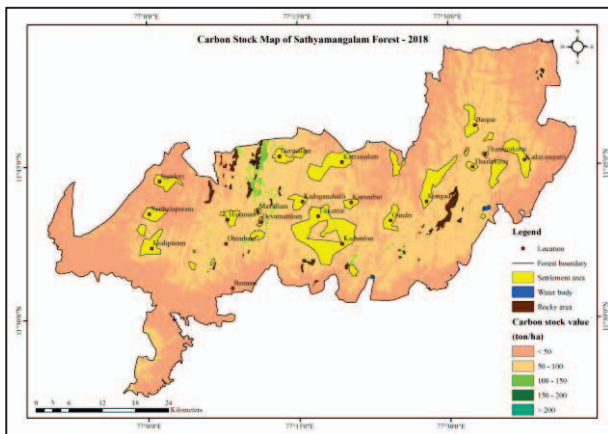


Figure 4. Carbon Stock map – 2018

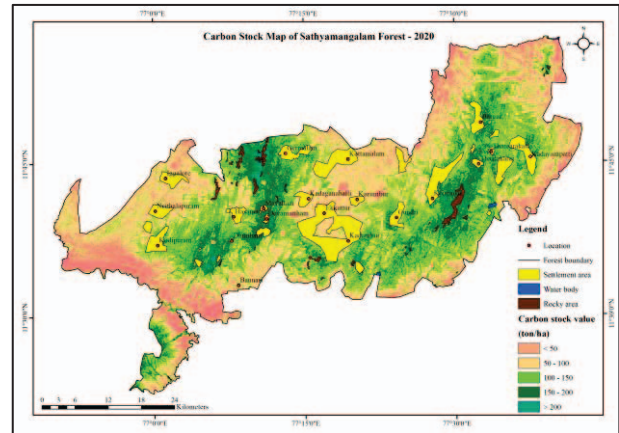


Figure 5. Carbon Stock map – 2020

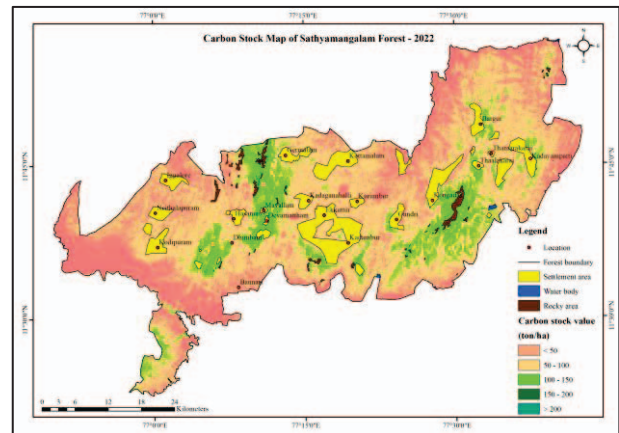


Figure 6. Carbon Stock map – 2022

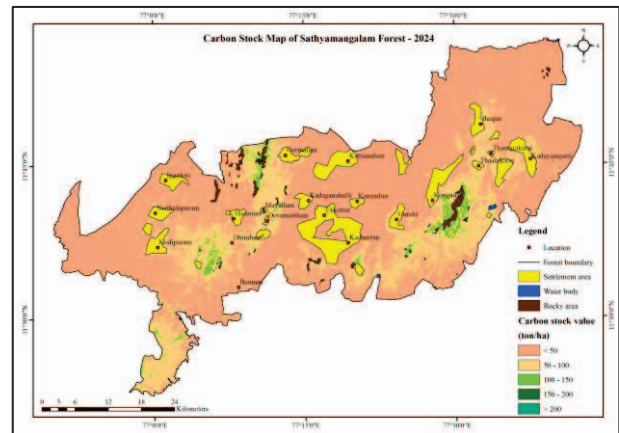


Figure 7. Carbon Stock map – 2024

6. CONCLUSION

The detection of forest carbon stock was a quite challenge in earlier times. In recent days, Geospatial

technology is adopted to estimate carbon stock more precise and economical. The changes occurred in carbon stock over the years due to various environmental actions. The factors that promote vegetation growth will also promote carbon storage in forests, which will increase the amount of carbon credits earned and raise the GDP of our nation. Finally, the present study states the continuous assessment and observation of carbon stock is predominant in and around Sathyamangalam forest.

7. ACKNOWLEDGEMENT

The authors would like to thank the SRM Institute of Science and Technology Kattankulathur for providing the vital facilities and encouragement to do the research.

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